Proceedings of the ASME 2026
International Mechanical Engineering Congress and Exposition
IMECE2026
November 8–12, 2026, Vancouver, Canada

IMECE2026-191477

HUMAN-IN-THE-LOOP ROBOTIC INSULATOR CLEANING WITH RGB-D ALIGNMENT AND CONTACT-FORCE-BASED AUTONOMOUS-TO-HAPTIC TRANSITION

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ABSTRACT

This paper presents a contact-aware dual-mode robotic framework for overhead-line insulator cleaning that extends our previous vision-based pose estimation and control approaches. An RGB-D perception pipeline is developed in which electrical insulators are segmented using a YOLOv26 Nano network and processed using geometric reasoning on the associated point cloud. Building on earlier vision-based alignment methods, pose estimation is enhanced through refined tangential plane estimation and refined orthogonal axes to improve pose stability for cylindrical symmetric objects. The resulting 6D pose is represented as a homogeneous transformation and used for autonomous end-effector alignment. The system is implemented on a Franka Emika manipulator equipped with RGB-D and force/torque sensing. Experimental evaluation is conducted in a laboratory-scale setup. Pose accuracy is assessed using a geometric overlap metric, achieving approximately 87% overlap. The robot autonomously approaches the insulator and transitions to haptic teleoperation upon force-based contact detection, demonstrating reliable perception, stable pose estimation, and safe autonomous-to-teleoperation handover.

Keywords: Robotic insulator cleaning, 6D pose estimation, RGB-D perception, force-based control, haptic teleoperation, human-in-the-loop robotics

1. INTRODUCTION

Robotic systems have been increasingly explored for electrical insulator inspection and maintenance in overhead power transmission infrastructure to reduce operator risk and improve operational efficiency. Insulator cleaning is a contact-rich task that requires accurate perception, precise alignment, and controlled physical interaction. In particular, reliable estimation of the insulator pose and safe interaction during contact are critical to prevent damage to both the infrastructure and the robotic system.

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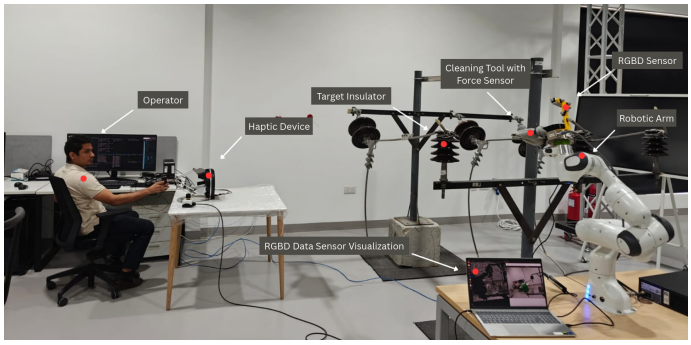


FIGURE 1: Complete robotic system setup for insulator cleaning, illustrating the dual-mode control framework with autonomous operation and transition to haptic teleoperation.

Fig. 1 illustrates the robotic system setup considered in this work, highlighting a dual-mode control framework that combines autonomous motion with haptic teleoperation. While prior research has demonstrated vision-based autonomous alignment and teleoperated manipulation for insulator cleaning, safely handling the transition between autonomous motion and operator-controlled physical interaction remains a key challenge due to sensing uncertainty and the risk of excessive contact forces.

In our earlier work, automatic tool alignment using eye-in-hand vision was proposed to estimate the insulator pose and autonomously align the robotic end effector [1]. This approach was later extended to a dual-mode framework combining autonomous motion and haptic teleoperation based on vision-derived 6D pose estimation [2]. However, in these approaches, the transition to teleoperation occurred at a predefined distance from the insulator, requiring additional operator involvement before physical contact and limiting operational efficiency.

To address this limitation, the present work introduces a contact-aware dual-mode robotic system in which the robot autonomously approaches the insulator until physical contact is detected through force sensing, triggering an automatic transition to haptic teleoperation. This strategy reduces operator workload

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while improving safety during contact-rich interaction.

The main contributions of this work are summarized as follows:

- An extension of prior vision-based 6D insulator pose estimation by incorporating refined tangential plane estimation and refined orthogonal axes for more stable and reliable pose recovery from RGB-D data.
- A contact-aware dual-mode control framework that enables automatic transition from autonomous motion to haptic teleoperation based on force feedback at physical contact.
- Experimental validation on a laboratory-scale insulator setup demonstrating stable pose estimation, reliable force-triggered control transition, and reduced operator workload compared to prior approaches.

The remainder of this paper is organized as follows. Section 2 reviews related work in the field. Section 3 describes the proposed 6D pose estimation method. Section 4 presents the robotic system, hardware integration, and experimental workflow. Finally, Section 5 concludes the paper.

2. RELATED WORK

2.1. Robotic Insulator Cleaning Systems

Robotic systems for overhead-line insulator cleaning have evolved from purely teleoperated mechanisms to semi-autonomous platforms, primarily to reduce operator risk during live-line maintenance. Early work focused on mechanical designs for insulator traversal and dry cleaning, such as the live-line cleaning robot with automatic string movement [3]. More recent systems integrate aerial platforms for reach, with fully autonomous UAVs performing insulator detection, dirty region segmentation, and cleaning trajectory generation [4]. Grounded manipulators have also advanced, with drone-robot hybrids using depth cameras for safe-distance cleaning of insulator chains [5]. These approaches highlight the need for reliable perception and controlled contact, though they often rely on distance-based safety triggers rather than direct force feedback.

However, these systems typically do not explicitly estimate object pose for precise alignment or incorporate contact-driven transitions, limiting their effectiveness in tasks requiring accurate physical interaction with the insulator surface.

2.2. Perception for Insulator Detection and Segmentation

Modern insulator maintenance systems leverage deep learning for robust detection under cluttered backgrounds and varying lighting. YOLO-family models have been adapted for insulator tasks, with ML-YOLOv5 achieving real-time defect detection on transmission lines [6], and Insu-YOLO using multiscale feature fusion for small-target insulator identification [7]. The latest version in YOLO series is YOLOv26[8] which have faster inference time and better results compared to previous versions. YOLOv26 provide a foundation for our RGB-D segmentation pipeline, which extends 2D detection with point-cloud processing for 6D pose recovery.

While effective for 2D tasks, these approaches lack the 3D geometric reasoning required for precise alignment, motivating the integration of depth-based reasoning in our pipeline.

2.3. 6D Pose Estimation from RGB-D Data

Accurate 6D pose estimation is critical for autonomous alignment in contact-rich tasks, particularly for symmetric cylindrical objects like insulators. Classical geometric methods such as RANSAC for model fitting [9] and PCA for principal axes [10] remain relevant for point-cloud refinement. Recent RGB-D approaches enhance this with deep features and hybrid reasoning, as seen in enhanced feature extraction pipelines [11]. Our prior vision-based alignment built on eye-in-hand pose estimation [1], which we extend here with refined tangential plane and orthogonal axes estimation for improved stability on cylindrical geometries.

Unlike prior work, which often assumes general object geometries, our method explicitly exploits the cylindrical structure of insulators, enabling more stable and consistent pose estimation despite inherent rotational symmetry.

2.4. Dual-Mode Control and Contact-Aware Teleoperation

Dual-mode frameworks combining autonomy and teleoperation address the limitations of fully manual or fully autonomous systems in uncertain environments. Our earlier dual-mode system transitioned to haptic teleoperation at a fixed distance using 6D pose [2], requiring premature operator intervention. Contact-aware control mitigates this by enabling autonomous approach until physical interaction, with force feedback ensuring safe transitions [12, 13]. Prototype teleoperated systems for insulator cleaning demonstrate force/6D pose integration [2], while general contact-rich manipulation emphasizes stable haptic handover [13]. This work advances these by triggering teleoperation directly at force-detected contact, implemented within ROS [14].

In contrast to prior systems that rely on distance-based switching, our approach enables a more natural and task-relevant transition by using force feedback to trigger teleoperation precisely at contact, improving both safety and operational efficiency.

3. 6D POSE ESTIMATION

This section describes the proposed approach for estimating the six-degree-of-freedom (6-DOF) pose of an electrical insulator using RGB-D data. Given an RGB image and its corresponding organized point cloud, the target insulator is first detected and segmented using a trained YOLOv26 Nano [8] model deployed as a ROS [14] service. In scenes containing multiple insulators, the instance with the largest segmented area is selected to ensure robust and unambiguous pose estimation.

Using the segmentation mask, the corresponding 3D points are extracted from the organized point cloud and processed through a geometric pipeline. The pipeline consists of three main stages: (i) estimation of a representative surface plane followed by normal refinement to obtain a *refined tangential plane*, (ii) orthogonal projection of the segmented points onto the refined tangential plane, and (iii) computation of *refined orthogonal axes* to recover a stable object orientation. Based on these components, the final insulator pose is represented as a 4×4 homogeneous

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transformation matrix defined in the camera coordinate frame. An overview of the complete pose estimation pipeline is shown in Fig. 2.

3.1. Input

RGB images and synchronized depth measurements are acquired using an Intel RealSense D455 RGB-D camera. Depth information is represented as an organized point cloud, where each image pixel (u, v) corresponds to a unique 3D point $\mathbf{p} = (x, y, z)$ in the camera coordinate frame. This one-to-one correspondence enables direct mapping from 2D segmentation masks to the corresponding 3D geometry of the insulator.

3.2. YOLOv26 Segmentation

The first step of the pipeline is semantic segmentation of the electrical insulator in the RGB image using a YOLOv26 Nano segmentation network trained on manually annotated images. The resulting segmentation mask is aligned with the organized point cloud. All pixels with non-zero mask values are selected, and their corresponding 3D points are extracted. Points with invalid depth values are discarded. The resulting 3D point set represents the segmented insulator and serves as input to subsequent geometric processing.

3.3. RANSAC Plane (Blue) vs. Refined Tangential Plane (Orange)

Let $\mathcal{P}$ denote the segmented point set. Since $\mathcal{P}$ may contain points from both the visible front surface and partial side surfaces of the insulator, a preprocessing step is applied to emphasize the frontal region. All points are projected onto the camera viewing direction $\mathbf{v} = (0, 0, 1)$, and the top 30% of points are retained to form the frontal point set $\mathcal{F}$ (with a minimum of 100 points to ensure numerical stability).

A planar surface is first fitted to $\mathcal{F}$ using the RANSAC algorithm [9],

$$ax + by + cz + d = 0, \tag{1}$$

where $\mathbf{n} = (a, b, c)$ denotes the plane normal. The normal is constrained such that $n_z < 0$ to ensure a camera-facing orientation. Due to the curved geometry of the insulator, the initial RANSAC estimate may deviate from the true tangential plane of the visible surface. To address this limitation, the plane normal is refined through a constrained pitch-angle search that minimizes penetration of the plane into the point cloud, resulting in a refined tangential plane, as summarized in Algorithm 1.

Algorithm 1 Normal Refinement via Pitch Search

Require: Initial normal $\mathbf{n}_0$, point set $\mathcal{P}$, centroid $\mathbf{c}$
Ensure: Refined normal $\mathbf{n}^*$
1: Generate candidate normals by rotating $\mathbf{n}_0$ about the x-axis: $\theta \in [-30°, +30°]$, step 2°
2: **for** each candidate normal $\mathbf{n}_i$ **do**
3: Enforce camera-facing constraint: $n_{i,z} < 0$
4: Compute penetration score: $s_i = \max_{\mathbf{p}\in\mathcal{P}}(\mathbf{p} - \mathbf{c}) \cdot \mathbf{n}_i$
5: **end for**
6: $\mathbf{n}^* \leftarrow \arg\min_i s_i$

After refining the plane normal, a robust plane center $\mathbf{c}^*$ is estimated by projecting the segmented insulator points along the refined normal $\mathbf{n}^*$ and computing the mean of the selected projected points. The initial RANSAC plane and the refined tangential plane are illustrated in Fig. 2.

3.4. Point Projection

All segmented 3D points are orthogonally projected onto the refined tangential plane to remove depth variation along the surface-normal direction:

$$\mathbf{p}_{\text{proj}} = \mathbf{p} - ((\mathbf{p} - \mathbf{c}^*) \cdot \mathbf{n}^*)\, \mathbf{n}^*. \tag{2}$$

This step produces a planar point distribution that preserves the spatial layout of the insulator surface and enables robust orientation estimation. The original insulator points are shown in green, while the projected points are shown in red in Fig. 2.

3.5. PCA (Yellow) vs. Refined Orthogonal Axes (Pink)

Principal Component Analysis (PCA) [10] is applied to the projected points $\mathcal{P}_{\text{proj}}$ to obtain an initial estimate of the dominant in-plane orientation axes. Since PCA axes are variance-driven, they may not align with the true geometric axes of the insulator and can appear diagonally oriented. Therefore, an in-plane refinement is performed to obtain refined orthogonal axes by rotating the axes about the plane normal $\mathbf{n}^*$ and selecting the rotation that minimizes the area of a bounding rectangle enclosing the projected points [15], as described in Algorithm 2. In Fig. 2, the initial PCA axes are shown in yellow, while the refined orthogonal axes are shown in pink.

Algorithm 2 PCA Axis Refinement via Bounding Rectangle Minimization

Require: Initial axes $(\mathbf{e}_1, \mathbf{e}_2)$, projected points $\mathcal{P}_{\text{proj}}$, plane normal $\mathbf{n}^*$
Ensure: Refined axes $(\mathbf{e}_1^*, \mathbf{e}_2^*)$
1: **for** $\theta \in [-20°, +20°]$ **do**
2: Rotate axes about $\mathbf{n}^*$ by θ
3: Project points onto the rotated axes
4: Compute bounding rectangle area $A(\theta)$
5: **end for**
6: Select $\theta^* = \arg\min_\theta A(\theta)$

3.5.1. Final Pose

The final 6-DOF pose of the insulator is represented as a homogeneous transformation,

$$\mathbf{T} = \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 & \mathbf{t} \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

where $\mathbf{r}_1$ and $\mathbf{r}_2$ are the refined in-plane orthogonal axes, $\mathbf{r}_3 = \mathbf{r}_1 \times \mathbf{r}_2$ ensures a right-handed coordinate system, and $\mathbf{t}$ is the centroid of the projected points. This pose estimate is used for autonomous end-effector alignment and subsequent robotic interaction. The resulting pose and the fitted cylindrical model are shown in Fig. 3.

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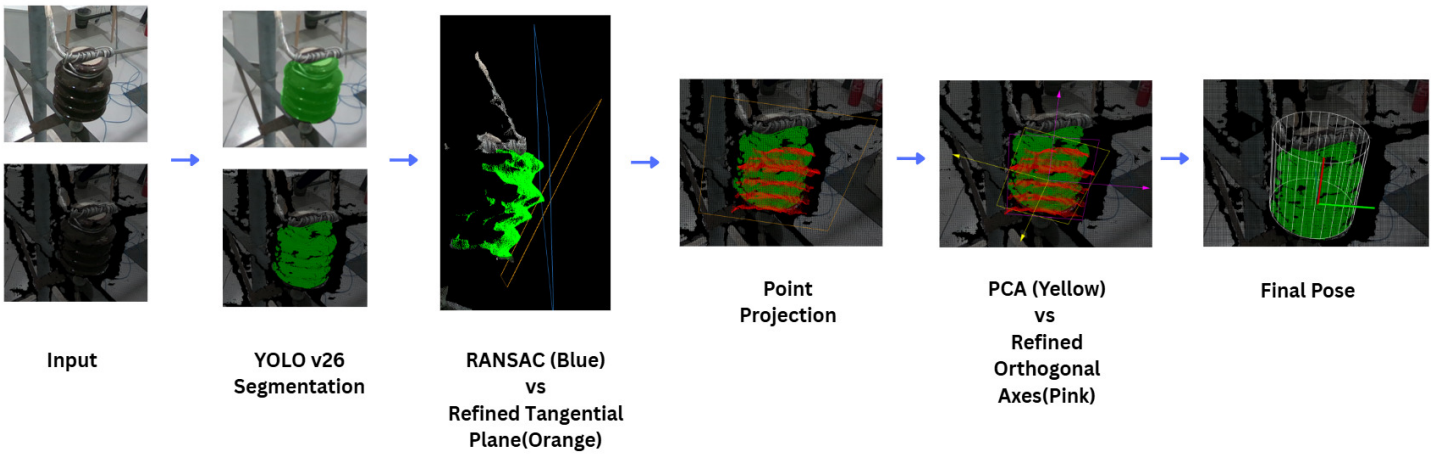


FIGURE 2: Overview of the proposed 6D pose estimation pipeline using RGB-D data.

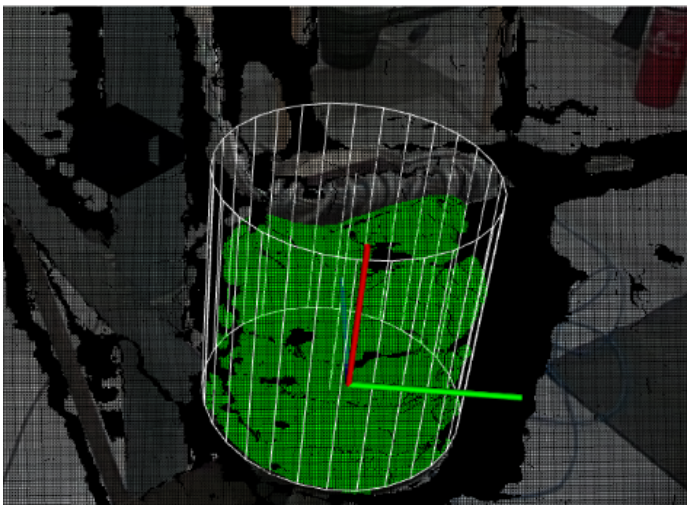


FIGURE 3: Estimated pose and cylinder on insulator points.

4. EXPERIMENTS

The experimental evaluation focuses on two key aspects of the proposed system. First, the accuracy and robustness of the vision-based 6D pose estimation pipeline—based on a refined tangential plane and refined orthogonal axes—is evaluated. Second, the complete robotic pipeline is validated, including dual-mode operation and the contact-aware, force-triggered transition from autonomous control to haptic teleoperation.

4.1. Experimental Setup

The experimental setup consists of a laboratory-scale dummy overhead-line configuration in which electrical insulators are mounted in a non-energized state. A 7-degree-of-freedom Franka Emika robotic manipulator is used, equipped with an end-effector-mounted RGB-D camera for perception and a wrist-mounted force/torque sensor for interaction sensing. This setup enables evaluation of both perception accuracy and physical interaction behavior in a controlled environment.

To evaluate the pose estimation module, an RGB-D dataset was collected containing electrical insulators observed from multiple viewpoints. The dataset includes variations in orientation as well as partial occlusions. For each RGB image, a corresponding organized point cloud is available, enabling direct association between image pixels and 3D spatial coordinates as required by the proposed pipeline. Ground-truth poses were obtained manually by interactively aligning a cylindrical geometric model with the insulator point cloud using a 3D visualization tool. The resulting poses were stored as 4×4 homogeneous transformation matrices and used as reference for qualitative evaluation and visual comparison.

The complete robotic pipeline was then evaluated on the same laboratory setup. The robot operates in a dual-mode control framework. In the first stage, an autonomous vision-guided mode is used, during which the end effector is aligned with the target insulator based on the pose estimated from the refined tangential plane and refined orthogonal axes. After alignment, the robot advances toward the insulator while interaction forces are continuously monitored. Upon physical contact with the insulator, detected through force sensing, the system automatically transitions to operator-guided teleoperation using a haptic interface. The operator then performs the cleaning task with force feedback. The overall robotic pipeline is illustrated in Fig. 4.

4.2. Segmentation Performance

The YOLOv26 Nano [8] model was used for electrical insulator segmentation as described in the pose estimation pipeline. The model was trained on the RGB-D dataset generated using the laboratory setup. All training images were manually annotated using tight polygon masks created with the LabelMe annotation tool [16]. The dataset was split into training and validation sets with an 80:20 ratio, and the model was trained for 500 epochs.

Evaluation on the validation set demonstrates strong segmentation performance. The trained model achieved a precision of 0.9541, a recall of 0.9033, and a mean average precision of 0.969 at an IoU threshold of 0.5 (mAP@50). The mean average precision across IoU thresholds from 0.5 to 0.95 (mAP@50–95) was 0.906, indicating consistent segmentation quality under varying overlap criteria.

The trained model reliably detects and segments multiple

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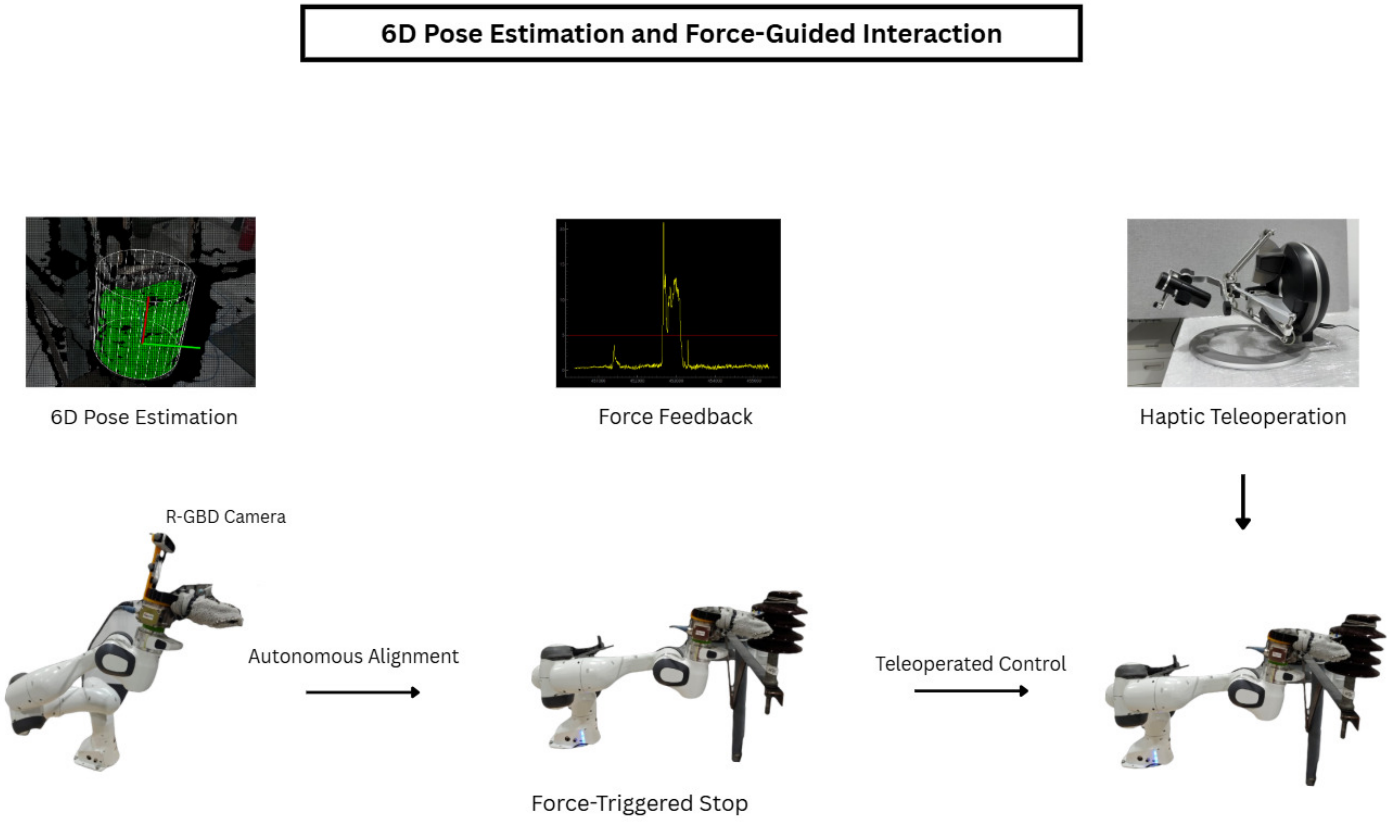


FIGURE 4: Overall system illustrating 6D pose estimation, autonomous alignment, and force-triggered transition to teleoperation.

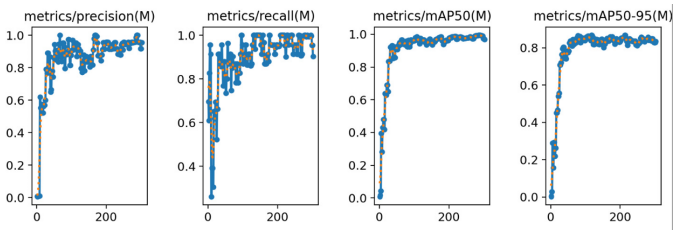
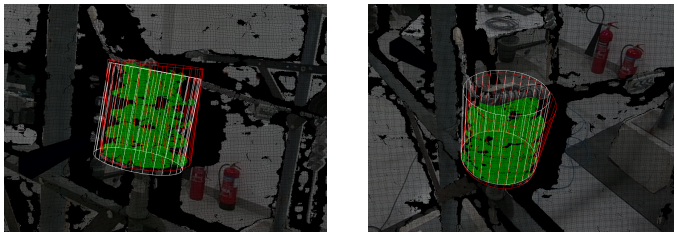


FIGURE 5: YOLOv26 Nano performance over training epochs.



(a) Frame 1 **(b) Frame 2**

FIGURE 6: Comparison of ground truth (red) and estimated (white) cylinder poses across two frames.

insulators within a single scene. For downstream processing, the largest detected instance is selected, as assumed in the pose estimation pipeline. The segmentation performance remains robust under changes in viewpoint, object orientation, and background clutter, providing reliable input for 3D point cloud extraction and subsequent geometric processing.

4.3. Pose Estimation Evaluation

The proposed 6D pose estimation pipeline was evaluated qualitatively by comparing the estimated pose with the manually annotated ground-truth pose. Since the insulator geometry can be approximated as a cylindrical object and is symmetric about its longitudinal axis, direct numerical comparison of orientation is not effective. Instead, the evaluation was performed using a geometric overlap metric.

Figure 6 shows a representative example in which the manually generated ground-truth cylindrical model (red) is compared with the cylindrical model estimated using the proposed approach (white).

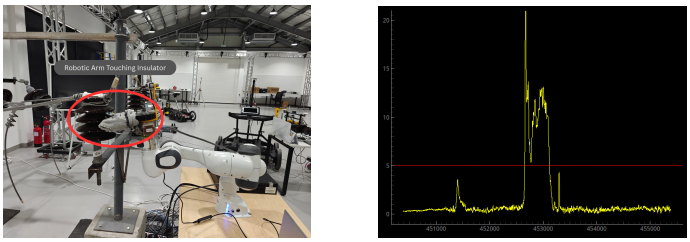
The overlap ratio is defined as the intersection volume divided by the volume of the ground-truth cylinder.

$$\text{Overlap Ratio} = \frac{V_{\text{intersection}}}{V_{\text{GT}}} \tag{3}$$

Across 50 samples, the average overlap ratio is approximately 87%, indicating strong agreement between the estimated and reference geometries.

It is important to note that the ground-truth annotations are manually generated and therefore subject to human error. In particular, small deviations in orientation or translation during annotation can lead to noticeable changes in the overlap ratio, as volumetric overlap is sensitive to angular misalignment. Conse-

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(a) Contact event (b) Force response

FIGURE 7: Force-triggered contact detection during insulator interaction: (a) end effector in contact with the insulator and (b) corresponding force signal measured at the wrist.

quently, the reported overlap ratios likely represent a conservative estimate of the true performance.

Further visual inspection confirms that the refined tangential plane provides a stable approximation of the visible insulator surface, while the refined orthogonal axes effectively correct the diagonal and unstable orientations produced by the initial PCA axes. The recovered pose accurately represents both the position and orientation of the insulator, and the fitted cylindrical model consistently overlays the corresponding point cloud. These results demonstrate the robustness and reliability of the proposed geometric refinement steps for downstream robotic alignment and interaction.

4.4. Force-Triggered Transition

During the autonomous approach phase, interaction forces were continuously monitored using the wrist-mounted force/torque sensor, as illustrated in Fig. 7. A predefined force threshold, determined empirically, was used to detect physical contact between the end effector and the insulator.

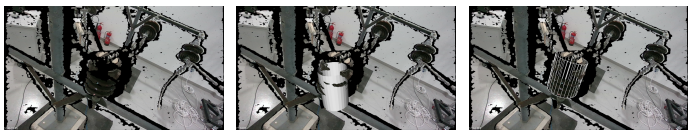
Once the measured interaction force exceeded this threshold, autonomous motion was immediately halted and control was automatically transferred to the operator through the haptic interface. This behavior was observed consistently across trials, with distinct force spikes corresponding to first-contact events.

The contact-aware, force-triggered transition provides a reliable and safe mechanism for switching from autonomous motion to teleoperated control. By allowing the robot to autonomously approach the insulator up to physical contact and enabling a smooth handover to operator control, the proposed strategy prevents excessive contact forces while preserving task continuity.

4.5. Performance Comparison

To evaluate the effectiveness of the proposed approach, its performance was compared qualitatively with the previous pose estimation and control framework described in our earlier work [2].

As shown in Fig. 8b, the previous method produces a cylindrical model that is tilted toward the supporting pole, indicating orientation bias. In contrast, the proposed method (Fig. 8c) produces a cylindrical model whose orientation is derived from the refined tangential plane and refined orthogonal axes, resulting in a geometrically consistent alignment approximately parallel to the pole.



(a) Raw point cloud (b) Previous method (c) Proposed method

FIGURE 8: Qualitative comparison of insulator pose estimation results: (a) raw point cloud, (b) pose estimated using the previous method [2], and (c) pose estimated using the proposed method.

TABLE 1: Qualitative comparison between previous and proposed systems

Property	Previous Approach	Proposed Approach
Accurate pose estimation	×	✓
Force-based contact detection	×	✓
Low operator workload	×	✓
Contact-aware autonomous-to-teleoperation transition	×	✓

In addition to improved pose estimation accuracy, the proposed system introduces a contact-aware control strategy. In the previous framework, the robot stopped at a predefined distance (approximately 35 cm) from the insulator, requiring multiple operator-controlled steps to reach contact and subsequently perform teleoperated insulator cleaning. In contrast, the proposed system autonomously proceeds until physical contact is detected and then transitions directly to haptic teleoperation for cleaning.

4.6. Discussion

The experimental results demonstrate that the proposed system achieves reliable perception and effective integration with robotic control. High segmentation accuracy ensures stable extraction of insulator geometry, while the refined tangential plane estimation and refined orthogonal axis selection produce consistent and repeatable pose estimates.

A key improvement over the previous approach lies in the resulting operational behavior. More accurate alignment enables the robot to approach the insulator more confidently, reducing the need for corrective teleoperation. The contact-aware, force-triggered transition further enhances usability by enabling a seamless handover from autonomous motion to human control at the moment of physical contact.

Overall, the results indicate that the proposed system improves both perception accuracy and operational efficiency, making it well suited for human-in-the-loop robotic inspection and maintenance tasks.

5. CONCLUSION

This paper presented a contact-aware dual-mode robotic framework for overhead-line insulator cleaning that extends prior vision-based pose estimation and control approaches. An enhanced RGB-D pose estimation method was introduced by incorporating refined tangential plane estimation and refined orthogonal axes, resulting in stable and reliable 6D pose recovery for cylindrical symmetric insulators. The estimated pose was used for autonomous end-effector alignment, while a force-based strategy enabled automatic transition from autonomous motion to haptic teleoperation upon physical contact.

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Experimental validation on a laboratory-scale setup demonstrated robust segmentation performance, consistent pose estimation, and smooth force-triggered handover to operator control. Compared to earlier approaches, the proposed system reduced operator workload and improved safety during contact-rich interaction. These results indicate that the framework is well suited for human-in-the-loop robotic inspection and maintenance tasks involving uncertain perception and physical contact.

AI USAGE DECLARATION

The authors acknowledge the use of AI-assisted tools during the preparation of this manuscript. These tools were employed to improve language clarity, grammar, and overall readability. All technical content, analysis, and results were independently developed and verified by the authors.

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